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Multi-Metric Evaluation of WRF Precipitation Schemes for Drought Mapping in Banyumas-Cilacap

Abstract:

Meteorological drought caused by regional climate anomalies like El Niño poses a serious threat to food security and water availability in monsoon-influenced regions such as the Banyumas and Cilacap regencies. This study evaluates the spatial and temporal performance of three Weather Research and Forecasting (WRF) model configurations using the Composite Performance Index (CPI) approach for drought potential mapping and affected area estimation at the sub-district level. The methods include multi-metric statistical integration, phase agreement analysis using the Standardized Precipitation Index (SPI), and spatial analysis based on the zonal histogram function. The results indicate that the DEF_SST scheme is the best-performing configuration, achieving an average CPI of 0.398 and maintaining Centroid Error below 18 km. Temporally, WRF-simulated rainfall patterns consistently follow the SPI-3 drought phases during the June-August (JJA) period, with rainfall reduction to 278.21 mm during the El Niño event of JJA 2015 (SPI-3 = -1.58). The Equal Weighting method produced a representative delineation of high-to-very-high drought vulnerability areas, concentrated in Kampunglaut, Kawunganten, and Cimanggu Sub-districts in Cilacap Regency, and Cilongok Sub-district in Banyumas Regency. These findings demonstrate that the integration of WRF modeling and Equal Weighting can support drought early warning systems and affected area estimation at the local scale.

Keywords: WRF, Meteorological Drought, SPI, CPI, El-Nino.

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1. Introduction

One of the most critical hydroclimatological factors for maintaining ecosystem stability is rainfall, particularly in tropical regions, where seasonal rainfall patterns heavily affect the availability of surface water [1]. Highly complex atmospheric dynamics influence changes in rainfall in Indonesia and, at the same time, impact the social and economic lives of communities. Meteorological drought is one of the effects of reduced rainfall. This phenomenon is defined as a condition in which precipitation falls below climatological norms over a specific period [2].

As a tropical archipelago, Indonesia routinely experiences meteorological droughts associated with the El Niño phenomenon. Java is a particularly vulnerable region because the island accounts for more than half of the country's rice production [3]. Reduced rainfall during the 2015-2016 El Niño period caused delays in the planting season and a decline in rice production across many regions of Indonesia; therefore, monitoring meteorological drought on Java is crucial for supporting national food security [4].

Two of the regions in Central Java most vulnerable to drought are the Banyumas and Cilacap regencies. During the 2015 El Niño event, the number of villages in Banyumas affected by drought increased from 14 to 48, spread across 17 subdistricts [5]. At the same time, Cilacap Regency experienced a significant decrease in rainfall compared to its normal climatological conditions [6]. These conditions indicate that the Banyumas-Cilacap region is highly sensitive to climate change, making it an intriguing subject for research on meteorological drought.

Meteorological drought is generally analyzed using the SPI developed by Thomas B. McKee and recommended by the World Meteorological Organization [7]. According to research conducted in Central Java, the SPI is capable of indicating spatial variations in drought and has a strong relationship with the El Niño-Southern Oscillation (ENSO) [8], [9]. However, the accuracy of the SPI is highly dependent on the quality of the precipitation data used, so its interpretation will be more meaningful if supported by adequate spatial distribution analysis.

Spatial approaches based on Geographic Information Systems (GIS) are also becoming increasingly popular in drought research because they are more efficient than analyses based on observation points [10], [11]. However, atmospheric model outputs for drought mapping at the district level remain very limited, and most previous studies still rely on observational data or satellite products. In fact, WRF model outputs can serve as an alternative precipitation data source, representing rainfall conditions more broadly and dynamically in drought analysis.

The WRF model has been widely used to simulate rainfall in tropical regions. Previous studies indicate that the selected physical parameterisation scheme significantly influences WRF performance [12], [13]. The WRF has also been used for drought studies in many countries, such as Spain and South Korea [14], [15]. However, most evaluations of rainfall models still focus on temporal statistical metrics, so aspects of spatial distribution and the movement of rain systems have not received much attention. In fact, the spatial accuracy of rainfall is crucial in drought analysis because errors in rainfall location can affect the results of estimates of affected areas. Therefore, a model with excellent temporal performance does not necessarily produce spatially accurate drought mapping.

2. Metodology

2.1. Study Area

The study area covers Banyumas and Cilacap regencies in Central Java Province. These two regions have distinct topographic characteristics, ranging from the volcanic slopes of Mount Slamet in the north to the coastal lowlands in the south, resulting in a complex rainfall distribution pattern [16]. Cilacap regency is dominated by coastal and lowland areas, while Banyumas features hilly terrain and volcanic slopes at higher elevations. These physical differences result in dynamic land-sea atmospheric interactions, particularly during the monsoon season and ENSO events, making the Banyumas-Cilacap region representative for evaluating the WRF model's ability to represent tropical precipitation in coastal areas and complex topographies [1]. The study location map is shown in Figure 1.

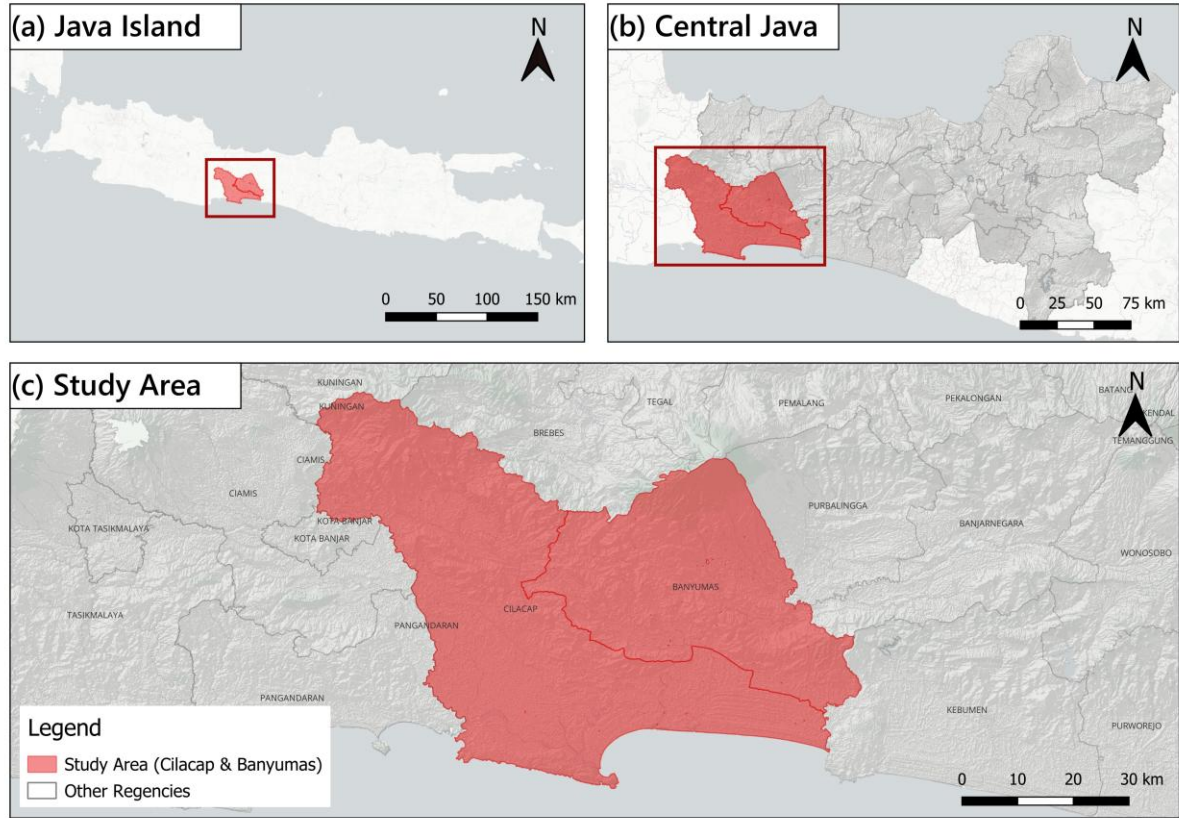


Fig. 1. Research Area

2.2 Data Used

This study uses model simulation data, observed rainfall data, gridded precipitation data, and administrative spatial data. The simulation data were obtained from the WRF model, which was run using three configuration experiments: DEF, PHY1, and DEF_SST. These three labels are experiment names used to distinguish the physical parameterization configurations and sea surface temperature (SST) treatments in the simulations. The DEF configuration represents the default physical configuration, PHY1 is an alternative physical configuration (physics configuration 1), while DEF_SST is the default WRF configuration with SST updates during the simulation. Simulations were conducted in the deepest domain with a spatial resolution of 10 km. The simulation was carried out for the June–August (JJA) period of three years, namely 2015, 2016, and 2017, selected to represent interannual rainfall variability associated with ENSO conditions in Central Java. Details of the physical parameterization configurations used in each experiment are presented in Table 1.

Table 1. Physical parameterization settings for each WRF simulation experiment.

Physical Parameterization	DEF	PHY1	DEF_SST
Microphysics	WSM3	Thompson	WSM3
Cumulus Scheme	Kain-Fritsch	Kain-Fritsch	Kain-Fritsch
Planetary Boundary Layer	YSU	YSU	YSU
Longwave Radiation	RRTM	RRTMG	RRTM
Shortwave Radiation	Dudhia	RRTMG	Dudhia
Land Surface Model	Noah LSM	Noah-MP LSM	Noah LSM

Rainfall observation data were obtained from 21 rain gauges operated by the Citanduy River Basin Agency, distributed across the Banyumas-Cilacap region. Data from 2015-2017 were used to evaluate model performance, while data from 2006-2024 were used in SPI calculations. This study also utilised Multi-Source Weighted-Ensemble Precipitation (MSWEP) version 3.16 data for spatial model validation. Additionally, administrative boundary data for Banyumas and Cilacap regencies in shapefile format were used in the spatial analysis process. A summary of the research data is presented in Table 2.

Table 2. Data used

Data Type	Source	Resolution/ Coverage	Period	Purpose
Simulation WRF (DEF, PHY1, DEF_SST)	WRF model output	10 km	2015, 2016, 2017	Analysis of the spatial distribution of rainfall and evaluation of model performance
Observed rainfall	BBWS Citanduy	21 station	2006- 2024	Validation of model simulation results and SPI calculations
MSWEP v3.16	(Wang et al., 2026)	0,1° / daily	2015- 2017	Spatial validation model
Administrative boundaries	(Indonesia Geospasial n.d.)	Shapefile	-	Clipping and zonal statistics

2.3 Methods

This study was conducted in several main stages, including data preprocessing, model performance evaluation using a multi-metric approach, determination of the optimal model configuration, analysis of meteorological

drought using the SPI, and spatial mapping of drought potential. The overall research workflow is shown in Figure 2.

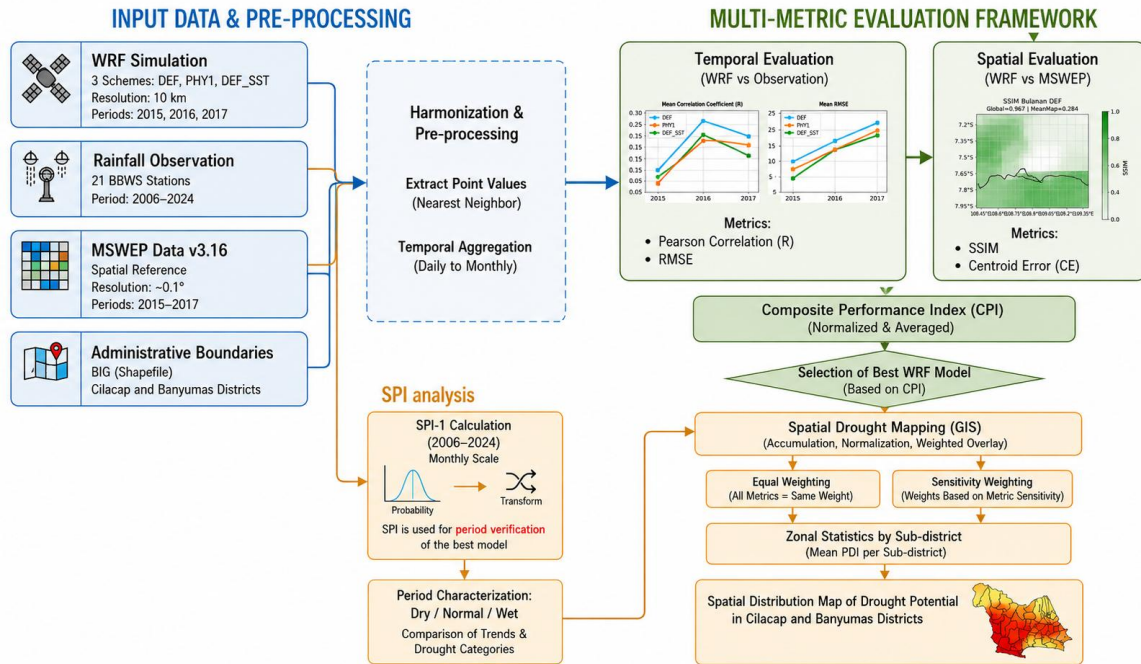


Fig 2. Research Phases

Pre-processing Data

The total precipitation from the WRF model output is obtained by summing the convective (RAINC) and non-convective (RAINNC) precipitation components shown in Equation (1):

$$P_{total} = RAINC + RAINNC \quad (1)$$

Rainfall values were then extracted from 21 observation stations using the nearest-neighbor method for temporal evaluation. The daily data were subsequently aggregated into monthly accumulations for spatial analysis and monthly totals for SPI calculations.

Multi-Metric Evaluation

The performance evaluation of the WRF was conducted using a multi-metric approach that included both temporal and spatial evaluations. The temporal evaluation utilized Pearson's correlation (R) and Root Mean Square Error (RMSE) based on daily observational data from 21 rain gauges. Pearson's correlation was calculated using Equation (2) [19]:

$$R = \frac{\sum_{i=1}^N (O_i - \bar{O})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^N (O_i - \bar{O})^2 \cdot \sum_{i=1}^N (S_i - \bar{S})^2}} \quad (2)$$

where O_i is the i -th observation, S_i is the i -th simulation, $\bar{O}$ denotes the mean of the observations, $\bar{S}$ denotes the mean of the simulations, and n is the number of observations.

Next, the RMSE is calculated using Equation (3) [19]:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (S_i - O_i)^2} \quad (3)$$

where S_i is the i -th simulation value, O_i is the i -th observed value, and n is the number of data points.

Spatial evaluation was performed by comparing the WRF rainfall distribution with that of the MSWEP using the Structural Similarity Index Measure (SSIM) and CE. SSIM was calculated using Equation (4) [20]:

$$\text{SSIM} = \frac{(2\mu_x\mu_y)(2\sigma_{xy})}{(\mu_x^2 + \mu_y^2)(\sigma_x^2 + \sigma_y^2)} \quad (4)$$

where μ_x and μ_y are the mean values of each image, σ_x^2 and σ_y^2 are the variances, and σ_{xy} is the covariance. The centroid of the rainfall distribution is calculated using Equation (5) [21]:

$$\bar{x} = \frac{\sum_i x_i I_i}{\sum_i I_i}, \bar{y} = \frac{\sum_i y_i I_i}{\sum_i I_i} \quad (5)$$

where x_i and y_i are the coordinates of the i -th grid cell, I_i represents the weight or intensity of rainfall at the i -th grid cell, and n is the number of grid cells. The centroid distance between WRF and MSWEP is calculated using the Haversine formula shown in Equations (6) and (7):

$$a = \sin^2\left(\frac{\Delta\varphi}{2}\right) + \cos\varphi_1 \cos\varphi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (6)$$

$$CE = 2R \arcsin(\sqrt{a}) \quad (7)$$

where R is the Earth's radius, ϕ_1 and ϕ_2 are the latitudes of the first and second points, λ_1 and λ_2 are the longitudes of the first and second points, and $\Delta\phi$ and $\Delta\lambda$ denote the differences in latitude and longitude.

All evaluation metrics are then integrated into a single composite index to facilitate comparisons between model configurations. Metrics with a “smaller is better” interpretation, such as RMSE and CE, are first normalized to the range 0-1 using Equation (8):

$$M_{norm} = \frac{M - M_{min}}{M_{max} - M_{min}} \quad (8)$$

where M_{min} is the minimum value, and M_{max} is the maximum value of all evaluation data. The CPI value is then calculated using Equation (9):

$$CPI = \frac{R + SSIM + (1 - RMSE_{norm}) + (1 - CE_{norm})}{4} \quad (9)$$

The scenario with the highest CPI value was selected as the best model for further analysis.

Calculation of the Standardized Precipitation Index (SPI)

Meteorological drought analysis was conducted using the monthly SPI (SPI-1) based on observed precipitation data from 2006 to 2024. Precipitation distribution was modeled using a two-parameter Gamma distribution according to Equation (10):

$$f(x; \alpha, \beta) = \frac{x^{\alpha-1} e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)}, x > 0 \quad (10)$$

where $\alpha > 0$ is the shape parameter, $\beta > 0$ is the scale parameter, and $\Gamma(\alpha)$ is the Gamma function. The distribution parameters are estimated using the maximum likelihood method shown in Equation (11) [22]:

$$\alpha = \frac{1}{4A} \left(1 + \sqrt{1 + \frac{4A}{3}} \right), \beta = \frac{\bar{x}}{\alpha} \quad (11)$$

where $A = \ln(\bar{x}) - \frac{1}{n} \sum_{i=1}^n \ln(x_i)$, where $\bar{x}$ is the monthly mean rainfall and n is the number of data points. Since the Gamma distribution is undefined for zero rainfall, the cumulative probability is corrected by incorporating the probability of zero rainfall using Equation (12):

$$H(x) = q + (1 - q)G(x) \quad (12)$$

where q is the probability of zero precipitation and $G(x)$ is the Gamma cumulative distribution function. The SPI value is then obtained by transforming it to a standard normal distribution using Equation (13):

$$SPI = \Phi^{-1}[H(x)] \quad (13)$$

where Φ^{-1} is the inverse of the standard normal cumulative distribution function. The SPI is used to identify dry, normal, and wet hydroclimatic conditions during the simulation period and to validate drought estimates from the best WRF configuration. The drought severity classification based on SPI values is shown in Table 3.

Table 3. Drought classification based on SPI values [22], [23]

SPI Score	Category
≥ 2.0	Very Wet
1.0 - 1.99	Wet
-0.99 - 0.99	Normal
-1.0 - -1.49	Dry
≤ -1.5	Very Dry

Spatial Mapping of Drought

The spatial distribution of precipitation from the best WRF configuration was analyzed using a GIS approach to identify potential drought vulnerability in the Banyumas-Cilacap region. Total precipitation accumulation was calculated using Equation (14):

$$P_{acc}(x, y) = \sum_{t=1}^n P(x, y, t) \quad (14)$$

The precipitation raster is then cropped to the administrative boundaries of the study area and normalized using Equation (15):

$$Z_{norm}(x, y) = 1 - \frac{P_{acc}(x, y) - P_{min}}{P_{max} - P_{min}} \quad (15)$$

where $P_{acc}(x, y)$ is the accumulated precipitation at pixel (x, y) , while P_{min} and P_{max} are the minimum and maximum precipitation values within the study

domain during the same period. To generate the Precipitation Deficit Index (PDI), the three rasters are combined using the weighted overlay method via Equation (16):

$$PDI(x,y) = w_k Z_k + w_n Z_n + w_b Z_b \quad (16)$$

where Z_k , Z_n , and Z_b are the Z_{norm} values for dry (k), normal (n), and wet (b) conditions, respectively, while w_k , w_n and w_b are the weights for each condition, subject to the constraint $\sum w = 1$. In the initial stage, the same weight was used for each condition ($w = 1/3$). PDI values range from 0 to 1, where values closer to 1 indicate a higher level of drought vulnerability. Subsequently, zonal statistics analysis was conducted at the subdistrict level to calculate the distribution and area coverage for each drought vulnerability class.

3. Result

Multi-Metric Evaluation of WRF Configuration Performance

The temporal performance of the WRF model was evaluated using the correlation coefficient (R) and RMSE. The results show variations in performance across model configurations and simulation periods. The evaluation results are presented in Figure 3.

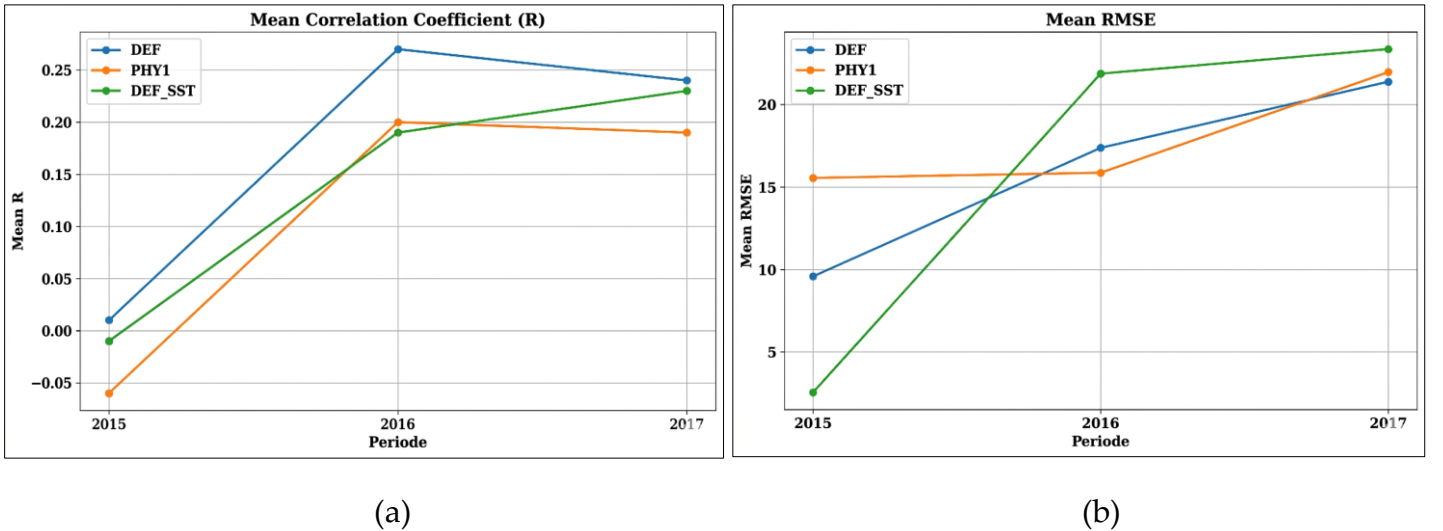


Fig 3. Temporal evaluation of model performance based on (a) the mean correlation coefficient and (b) the mean root mean square error (RMSE).

The results show that the DEF scheme has the highest correlation value compared to the other schemes, with an R value of approximately 0.27 in 2016,

which decreased slightly to 0.24 in 2017. On the other hand, the PHY1 and DEF_SST schemes also showed an increase in correlation, although their values remained below that of DEF. The error values for the RMSE parameter tended to increase annually across all schemes. In 2015, the lowest RMSE was observed in the DEF_SST scheme, while 2016-2017 showed a significant increase in error.

In the 2015 PHY1 scheme, the low correlation value ($R = -0.06$) does not necessarily indicate model ineffectiveness. This condition was more influenced by the fairly strong 2015 El Niño climate anomaly, which caused unusual rainfall patterns. The standard WRF parameterization, typically used for normal monsoon conditions, became more difficult to represent, and convection slowed due to SST mismatches in the Indo-Pacific region. Phase agreement between observational data and simulations also decreased because external ENSO factors influenced rainfall more than local atmospheric processes. Furthermore, the increase in the RMSE value from 2015 to 2017 must be understood proportionally because the RMSE is heavily influenced by rainfall amounts. During wet periods, higher rainfall amounts automatically increase the absolute difference between observations and simulations, so an increase in the RMSE does not necessarily indicate a decline in model performance.

In addition to temporal evaluation, model performance analysis was also conducted spatially by comparing the distribution of simulated WRF precipitation with MSWEP data using SSIM. SSIM was used to assess the degree of similarity in precipitation distribution patterns between the datasets. The SSIM results for each model configuration and simulation period are shown in Figure 4.

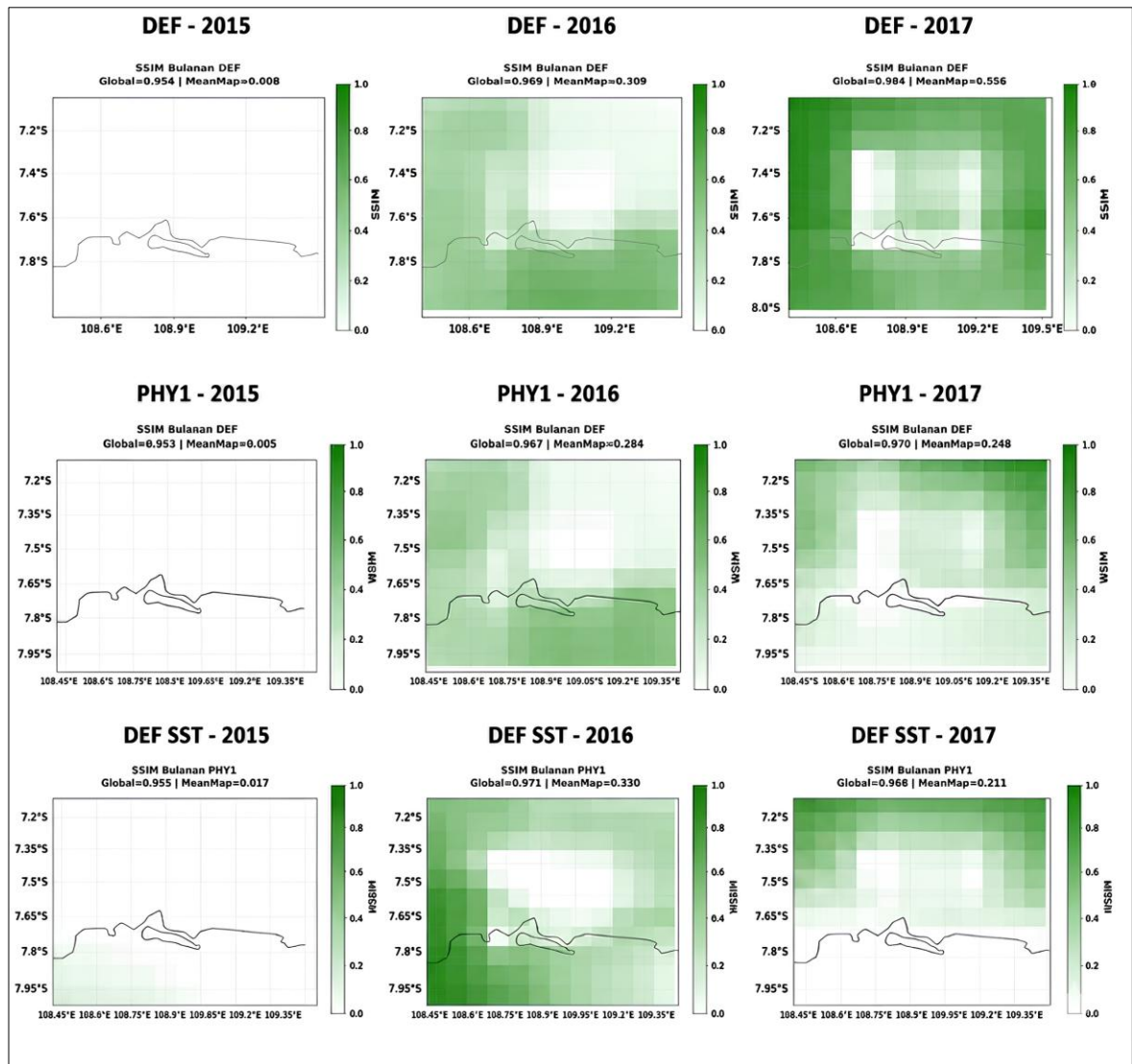


Fig 4. Spatial distribution of SSIM values from the simulation results for the DEF, PHY1, and DEF_SST scenarios.

Based on the distribution pattern of SSIM values in Figure 4, it is evident that the model's spatial similarity was still very low in 2015, as indicated by the predominance of values close to zero across all three scenarios. The SSIM values close to zero in 2015 do not entirely indicate a failure of the model but are rather influenced by extremely low rainfall conditions caused by El Niño. The spatial structure of precipitation became more difficult to compare because nearly the entire region experienced a uniform decrease in rainfall. Consequently, the SSIM values also decreased. Thus, this condition demonstrates the model's ability to consistently represent the dry phase. The DEF_SST scheme produced the highest SSIM values, averaging 0.56 during the wet year of 2017. These results indicate that the use of variable SST can

improve the representation of spatial rainfall distribution, particularly in the southern coastal area of Cilacap, which is influenced by ocean-atmosphere interactions. The model's ability to represent the center of rainfall distribution was further tested using the CE metric presented in Figure 5.

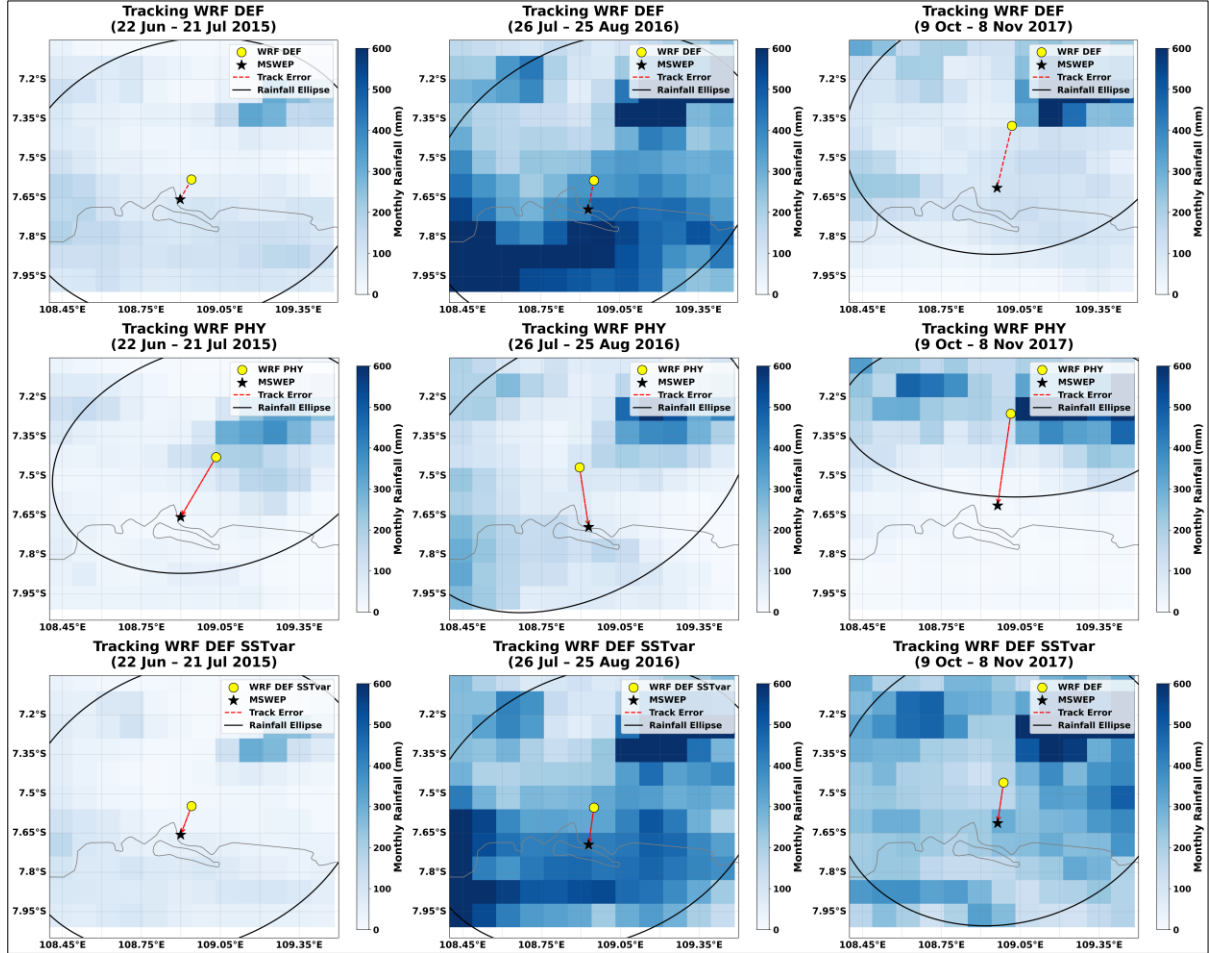


Fig 5. Spatial evaluation of the simulated rainfall distribution from the WRF model using the MSWEP method with CE for the DEF, PHY1, and DEF_SST configurations.

Differences in centroid positions across schemes and years indicate that the 10 km model resolution still has limitations in capturing the influence of small-scale topography, which contributes to the formation of convective clouds locally. In the wet year of 2017, the PHY1 scheme showed greater variation in centroid positions, with a centroid error of 39.31 km, while DEF_SST was only about 17.40 km, indicating that the location of atmospheric convergence became less stable. Conversely, the DEF_SST scheme was able to maintain a more consistent centroid position because the use of variable SST bottom

boundary conditions provided a better representation of air mass movement from the sea to the land in the Cilacap coastal region.

After temporal and spatial evaluations were conducted, all metrics were then integrated into a multi-metric approach to comprehensively assess the WRF model's performance. This approach was used to accommodate both temporal and spatial aspects simultaneously in the model evaluation process. A summary of the multi-metric evaluation results is presented in Table 4.

Table 4. Results of the WRF simulation performance evaluation for the 2015-2017 period for each configuration, based on Mean R, Mean RMSE, SSIM mean map, and Centroid Error.

Period	Scheme	Mean R	Mean RMSE	SSIM mean map	Centroid Error (km)
2015	DEF	0.01	9.58	0.004596644	9.677504381
2015	PHY1	-0.06	15.55	0.016779313	29.40026606
2015	DEF_SST	-0.01	2.55	0.00782935	12.96792066
2016	DEF	0.27	17.37	0.283620489	12.46251233
2016	PHY1	0.2	15.86	0.33001686	25.57764159
2016	DEF_SST	0.19	21.87	0.308911143	15.90057333
2017	DEF	0.24	21.38	0.247805319	27.03762045
2017	PHY1	0.19	21.96	0.210958126	39.30513991
2017	DEF_SST	0.23	23.36	0.555828569	17.40491191

Based on this data, the 2015 period presented the greatest challenge for all schemes, with correlation values close to zero or even negative for PHY1 (-0.06). However, the DEF scheme in 2015 had the smallest centroid error (9.68 km) and a relatively low RMSE value. The SST modification (DEF_SST) showed interesting consistency; the scheme successfully reduced the centroid error significantly to 17.40 km in 2017 and recorded the highest local structural similarity (SSIM_mean_map 0.55), indicating that, although the RMSE value increased slightly, improvements in sea surface boundary conditions provided better spatial stability.

The absence of a single scheme that consistently outperforms across all metrics highlights the importance of using CPI as an integrated evaluation approach. This situation illustrates a trade-off between temporal and spatial accuracy; schemes with high temporal correlation do not necessarily exhibit good spatial performance. In tropical coastal regions such as Banyumas-Cilacap, spatial aspects such as centroid position and rainfall distribution are critical factors in determining the reliability of drought analysis. Therefore, all

temporal and spatial evaluation metrics are integrated into the Composite Performance Index (CPI) to obtain a more comprehensive assessment of model performance. The results of the CPI integration are shown in Figure 6.

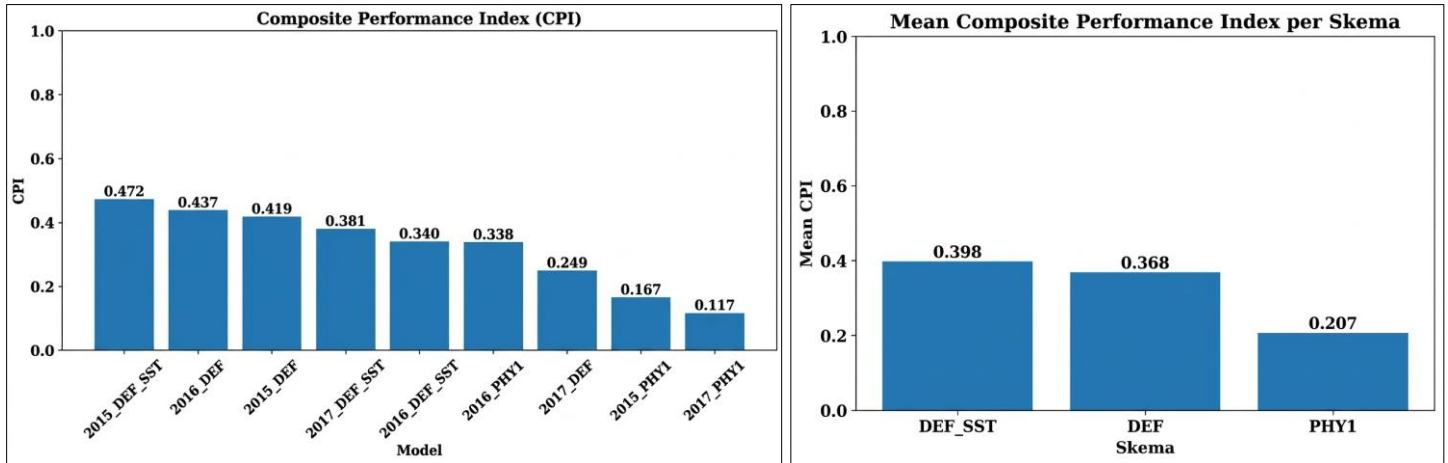


Fig 6. CPI graph for each combination of year and scheme.

According to these results, the DEF_SST configuration emerged as the best configuration with the highest average CPI value of 0.398, slightly outperforming the standard DEF scheme with a value of 0.368. The default WRF parameterization is actually quite suitable for the Banyumas-Cilacap region, as indicated by the relatively small difference in CPI values between DEF_SST and the standard DEF, which is approximately 0.030. The use of improved SST boundary conditions can enhance performance. This suggests that future research should focus more on the quality of SST data, such as using satellite SST products with higher resolution, rather than introducing changes to physical parameters that may degrade model performance, as seen in the PHY1 scheme. The consistently low performance of PHY1 throughout the year and the metrics indicate a fundamental mismatch between the physical parameterization and the atmospheric dynamics.

Hydroclimatological Characteristics of the Simulation Period Based on the SPI

A time series analysis of the SPI was conducted to examine changes in the region's hydroclimatological conditions over the past two decades. The results of the analysis indicate that climate variability differs across periods,

particularly in terms of wet and dry conditions. A visualization of these changes in hydroclimatological conditions is shown in Figure 7.

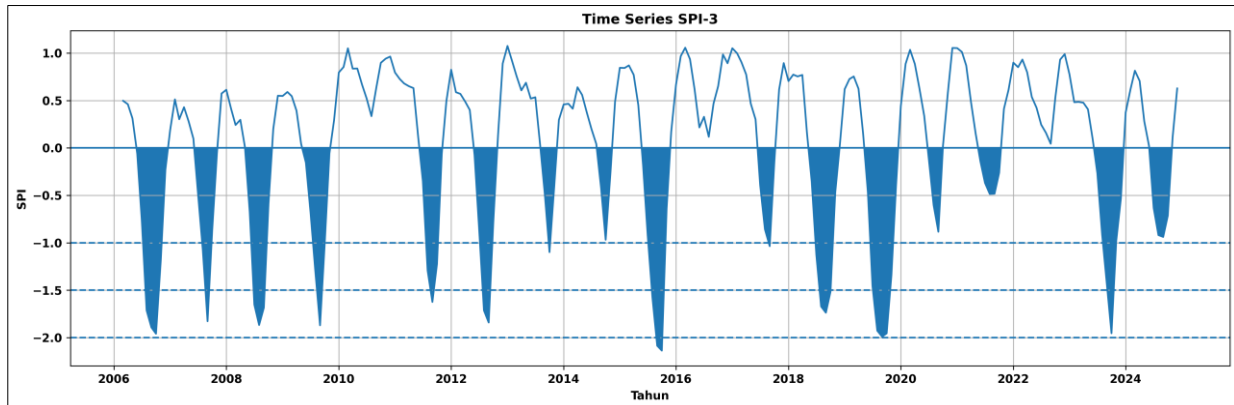


Fig 7. Time series of the SPI index for the Banyumas-Cilacap region for the period 2006-2024.

The figure shows fluctuations in wet-dry conditions in the study area during 2006–2024, with several negative SPI-3 anomalies reflecting recurring drought events. To ensure consistent year-to-year comparisons, the analysis focused on the June, July, August (JJA) period.

Based on this time series, the 2015 JJA was the driest period due to the influence of a strong El Niño, while the 2016 and 2017 JJAs were both classified as normal, with a slight tendency toward wetter and drier conditions, respectively. This pattern indicates that the impact of El Niño on reduced rainfall during the dry season is stronger than the increase in rainfall during La Niña. These results are consistent with Firmansyah et al. (2022) and form the basis for selecting the years 2015, 2016, and 2017 as representations of extreme dry conditions, normal with a tendency toward wetter conditions, and normal with a tendency toward drier conditions during the JJA period. A summary of the index characteristics for the months focused on in the simulation is presented in Table 5.

Table 5. SPI index values and categories for the months covered by the simulation.

Year	Month	SPI	Category SPI
2015	8	-1.578430374	Very Dry
2016	8	0.1169326448	Normal
2017	8	-0.8577640873	Normal

In August 2015, the study area experienced very dry conditions with an SPI-3 value of -1.58. In contrast, August 2016 and 2017 fell into the normal category with SPI-3 values of 0.12 and -0.86, respectively, indicating slightly wet and slightly dry trends. These variations in SPI-3 values represent different hydroclimatological conditions and were therefore used to evaluate the sensitivity of the WRF model by comparing them with the simulated cumulative precipitation shown in Figure 8.

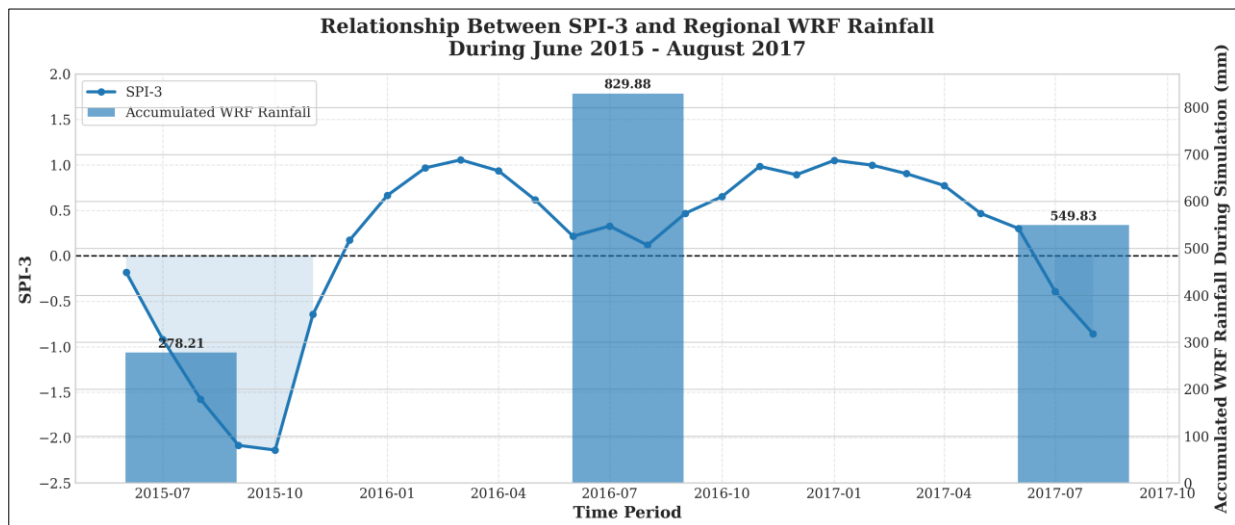


Fig 8. Graph showing the relationship between the SPI index and accumulated precipitation as simulated by the WRF model.

The graph shows a consistent relationship between SPI-3 and simulated WRF rainfall accumulation, particularly under conditions of extreme drought. When SPI-3 reached its lowest value in the 2015 JJA period (-1.58), the model also produced the lowest rainfall accumulation (278.21 mm). In the normal category, this relationship is not linear, as evidenced by the fact that rainfall in JJA 2016 (829.88 mm) was higher than in JJA 2017 (549.83 mm), even though both years fell within the normal SPI category. These findings indicate that the correlation between SPI-3 and rainfall is more pronounced under extremely dry conditions than under normal conditions.

Drought Potential Map Results

A spatial analysis was conducted to examine variations in the distribution of JJA-accumulated rainfall during each simulation period in the Banyumas-Cilacap region. The mapping results indicate differences in rainfall

distribution patterns across the three JJA simulation periods. The spatial distribution of simulated rainfall accumulation from the WRF model for the periods 2015, 2016, and 2017 is shown in Figure 9.

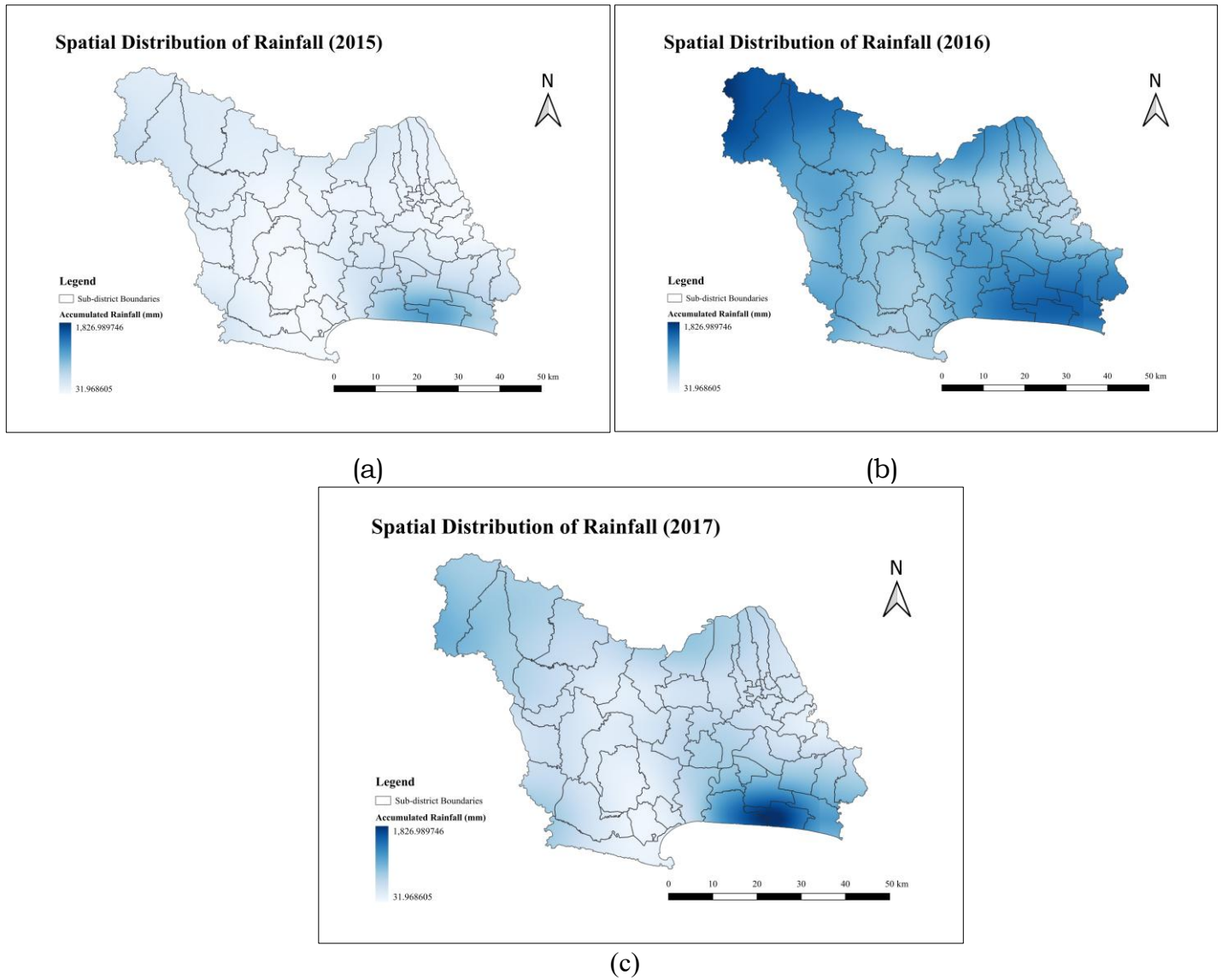


Fig 9. Spatial distribution of accumulated precipitation from WRF simulations for the periods (a) 2015, (b) 2016, and (c) 2017 in the Banyumas-Cilacap region.

Based on Figure 9, the JJA rainfall distribution shows a spatial pattern consistent with the results of the SPI-3 analysis and WRF rainfall accumulation. During the 2015 JJA, nearly the entire Banyumas-Cilacap region was dominated by low rainfall, reflecting a widespread rainfall deficit due to the influence of a strong El Niño. In contrast, the 2016 JJA showed a more widespread increase in rainfall, particularly in the northwest and east, although it was still classified as normal according to the SPI-3. This condition indicates

that the SPI, as a standardized index, does not always represent spatial variations in rainfall in detail. In the 2017 JJA, rainfall was again concentrated along the south–southeast coast, while other regions remained relatively dry, consistent with SPI-3 values that tended to be drier than in 2016. Overall, this spatial pattern reinforces the interpretation of the SPI-3.

Drought potential mapping was conducted using the equal weighting method, which assigns equal weights to each component parameter. The results of this spatial modeling are presented in Figure 10.

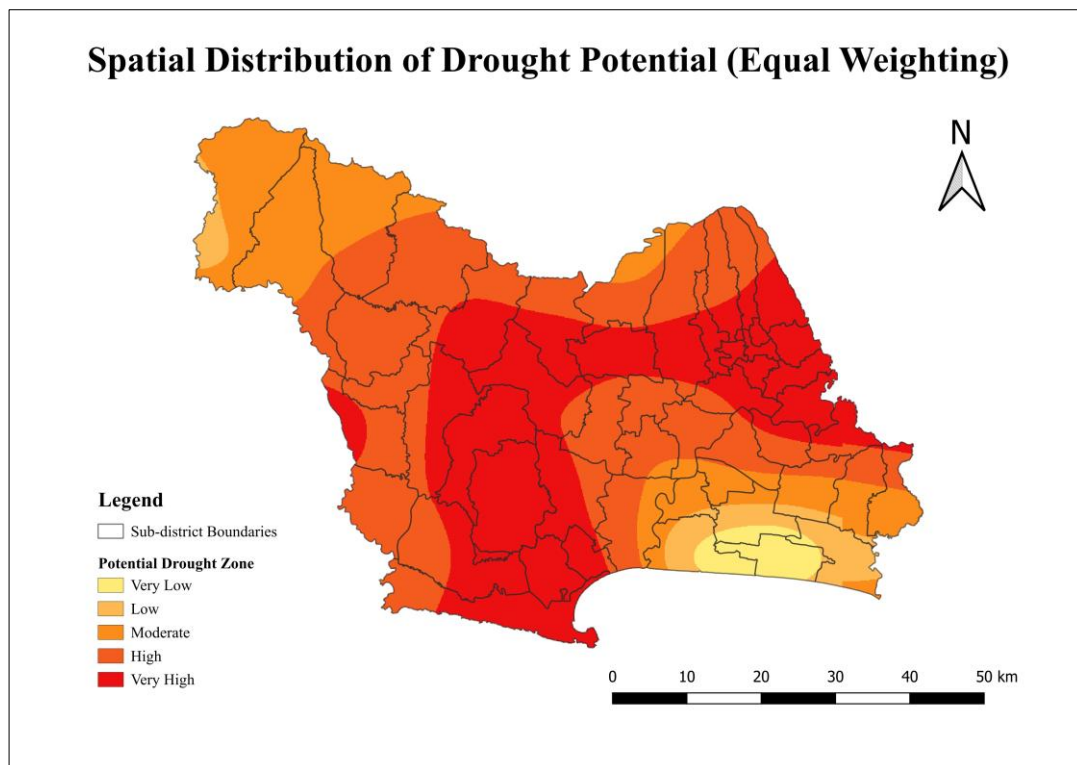


Fig 10. Drought Potential Map with Equal Weighting

Figure 10 shows the spatial distribution of the Precipitation Deficit Index (PDI) using the Equal Weighting method for the JJA periods of 2015, 2016, and 2017. In general, drought vulnerability levels form a gradient from north to south, with the very high zone dominating the central to southwestern regions as well as parts of the southern coast, while the very low to low zones appear only in limited areas around Binangun-Kroya and parts of the northwest. This pattern aligns with the distribution of previous JJA rainfall, which shows that the southern region consistently receives lower rainfall than other regions.

The dominance of the very high zone in the south is believed to be related to the region's physical characteristics, which make it more vulnerable to rainfall deficits, while the northern region benefits from the orographic influence of the Mount Slamet complex, which supports rainfall formation. Several subdistricts, such as Kawunganten, Kampung Laut, Bantarsari, Kedungreja, Patimuan, Gandrungmangu, and Jeruklegi, also fall into the very high category, which aligns with reports of clean water crises during the dry season. Conversely, the very low zone around Binangun–Kroya is believed to be influenced by the presence of an irrigation network and its proximity to the Serayu River, making it relatively more resilient to rainfall deficits.

A quantitative analysis of the spatial impacts of drought was conducted using the Zonal Histogram function at the subdistrict level. The calculations focused on the total area of high to very high vulnerability categories in square kilometers. A summary of the results is presented in Table 6.

Table 6. Summary of the area affected by drought in the High-Very High categories by subdistrict, based on the equal weighting method.

Regency	Subdistrict	Mean PDI	Dominant Class	High-Very High Area (km ²)
Cilacap	Cimanggu	0.781	High	137.64
Cilacap	Kampunglaut	0.916	Very High	133.55
Cilacap	Kawunganten	1	Very High	128.51
Banyumas	Cilongok	0.871	High	126.13
Cilacap	Karangpucung	0.915	Very High	124.56
Cilacap	Cilacap Selatan	0.938	Very High	123.25
Cilacap	Gandrungmangu	0.932	Very High	119.28
Cilacap	Bantarsari	1	Very High	109.8
Cilacap	Cipari	0.8	High	107.61
Banyumas	Lumbir	0.97	Very High	106.52
Cilacap	Jeruklegi	0.94	Very High	99.24
Banyumas	Gumelar	0.915	Very High	94.12
Cilacap	Kedungreja	0.863	High	81.8
Cilacap	Patimuan	0.8	High	78.5
Banyumas	Wangon	0.814	High	69.41
Banyumas	Ajibarang	0.99	Very High	68.73
Banyumas	Pekuncen	0.752	High	59.71
Banyumas	Sumbang	0.948	Very High	57.78
Cilacap	Kesugihan	0.729	High	57.6
Banyumas	Kedungbanteng	0.86	High	55.49
Cilacap	Cilacap Tengah	1	Very High	50.74
Cilacap	Sidareja	0.8	High	49.16
Banyumas	Patikraja	0.942	Very High	47.02
Banyumas	Baturraden	0.854	High	46.13
Banyumas	Rawalo	0.772	High	45.1
Banyumas	Kebasen	0.779	High	44.88

Regency	Subdistrict	Mean PDI	Dominant Class	High-Very High Area (km²)
Banyumas	Somagede	0.944	Very High	42.83
Banyumas	Banyumas	0.882	High	41.97
Banyumas	Jatilawang	0.774	High	41.9
Banyumas	Kalibagor	1	Very High	41.04
Banyumas	Purwojati	0.801	High	41.02
Cilacap	Cilacap Utara	0.979	Very High	29.59
Banyumas	Sokaraja	1	Very High	29.5
Banyumas	Karanglewas	0.951	Very High	29.44
Banyumas	Kembaran	1	Very High	27.29
Banyumas	Purwokerto Selatan	1	Very High	15.17
Banyumas	Purwokerto Timur	1	Very High	9.98
Banyumas	Purwokerto Barat	1	Very High	9.51
Banyumas	Purwokerto Utara	1	Very High	8.03

The results of the zonal histogram show that the areas classified as High-Very High is the largest in the subdistricts of Cimanggu (137.64 km²), Kampunglaut (133.55 km²), Kawunganten (128.51 km²), Cilongok (126.13 km²), and Karangpucung (124.56 km²). A comparison across regions shows that the average PDI value does not always correlate with the size of the affected area. Subdistricts in the Purwokerto urban area, such as East Purwokerto, West Purwokerto, North Purwokerto, South Purwokerto, Kembaran, Sokaraja, and Kalibagor, have the highest Mean PDI value (1,000), but the area classified as High-Very High is relatively limited (8-41 km²) due to the small size of their administrative territories. Conversely, subdistricts with larger administrative areas, such as Cimanggu and Cilongok, show a larger coverage of affected areas even though their average PDI values fall only within the High category. These findings suggest that setting priorities for drought mitigation should take into account the extent of the affected area in addition to the intensity level.

The subdistricts of Kampunglaut, Kawunganten, Karangpucung, Gandrungmangu, Bantarsari, Jeruklegi, Kedungreja, and Patimuan, which are predominantly classified as Very High were also reported to have experienced clean water crises during the dry season. This correspondence provides an initial indication that the modeling results accurately represent on-the-ground conditions, although they have not yet been verified through quantitative validation using historical drought data. On the other hand, the interpretation of the Very High category in the urban area of Purwokerto must be approached with caution, as meteorological rainfall deficits do not always correlate directly with community vulnerability levels, given that this area is supported by a

relatively more adequate clean water supply infrastructure compared to rural areas that still rely on surface water and shallow wells.

4. Discussion

Multi-metric integration using the CPI approach indicates that the modified sea surface temperature (SSTvar) scheme exhibits more stable spatial performance compared to other schemes. These results are consistent with the study by Pilatin et al. (2021), which noted that the use of time-varying SST data in the WRF model can reduce overestimation and improve rainfall distribution, particularly in coastal areas with complex topography. Conversely, the PHY1 scheme exhibits a centroid error of up to 39.31 km, indicating that an inappropriate combination of physical parameterizations can increase the uncertainty in the spatial distribution of rainfall [25].

PHY1's performance was consistently the lowest across all periods and evaluation metrics, with an average CPI of just 0. These results indicate a mismatch between the PHY1 physical parameterization combination and the atmospheric characteristics in the Banyumas-Cilacap region. PHY1 employs a combined scheme of the Thompson microphysics, RRTMG radiation, and a modified Noah-MP land surface model, which may lead to physical imbalances in the model. The use of the RRTMG radiation scheme in the WRF tends to produce higher precipitation frequencies in tropical regions, which can trigger excessive rainfall and amplify rainfall location errors [26]. On the microphysics side, replacing WSM3 with Thompson without local sensitivity testing also risks being less suitable for tropical coastal convection patterns, which tend to be shallow and unstable. Additionally, Noah-MP's performance is significantly influenced by local vegetation and soil characteristics [27]. The use of Noah-MP without calibration in areas with heterogeneous land cover is suspected to also affect atmospheric humidity estimates. Therefore, modifications to WRF parameterizations in tropical coastal regions should be implemented gradually to more clearly identify sources of model error.

Validation using the SPI-3 also indicates that the WRF model is quite sensitive to changes in regional climate conditions. During the El Niño event of JJA 2015 with an SPI-3 value of -1.58 (very dry), the simulated rainfall from

the WRF model decreased to 278.21 mm. These results support the finding that ENSO, particularly the El Niño phase, significantly influences the reduction in rainfall on Java Island [28]. Furthermore, the use of WRF-simulated rainfall standardized to a consistent seasonal (JJA) accumulation period is considered capable of producing a more objective identification of vulnerable areas, since isolating the interannual climate signal from seasonal cycle variability can reduce spatial bias and improve the accuracy of estimates of drought-prone areas [29].

Although the modeling results are able to depict general patterns of drought vulnerability, this study still has several limitations. One of the main limitations lies in the spatial resolution of the WRF model, which still uses a 10-km grid. This resolution causes a smoothing effect, making the model less capable of capturing local weather phenomena and complex topographic influences, particularly in coastal and hilly regions [30].

This resolution limitation also affects the model's accuracy in representing daily precipitation in detail. Additionally, the WRF output in this study still uses raw data without data assimilation or statistical bias correction [31]. The drought analysis conducted also remains focused on meteorological parameters, particularly precipitation, and has not yet considered land physical factors such as soil type and water retention capacity.

Therefore, future research is recommended to apply bias correction methods, such as quantile mapping, to improve the agreement between simulation results and observed data [32], [33]. The use of higher grid resolutions through dynamic downscaling techniques is also necessary to better represent local variations in coastal areas and complex topography [34]. Furthermore, integrating meteorological parameters with the physical characteristics of the environment is crucial for producing more comprehensive and practical drought risk maps to support disaster mitigation efforts [35].

5. Conclusion

This study has successfully evaluated the spatial and temporal performance of the WRF model and integrated it into drought potential mapping at the subdistrict level. Based on the CPI results, the DEF_SST configuration emerged

as the best scheme with an average value of 0.398, as it was able to maintain the centroid error below 18 km for each period. Temporally, the rainfall patterns from the WRF simulations also consistently align with drought conditions based on the SPI-3 during the June–August (JJA) period, as evidenced by a rainfall decrease to 278.21 mm during the JJA 2015 El Niño event with an SPI-3 value of -1.58.

Using the equal weighting method, the sub-districts of Cimanggu (137.64 km²) and Kawunganten (128.51 km²) in Cilacap Regency, along with Cilongok (126.13 km²) in Banyumas Regency, exhibit the most extensive high-to-very-high drought vulnerability areas according to the zonal histogram results. However, with the WRF's 10-km grid resolution, spatial generalization effects still exist. Consequently, to improve the accuracy of modeling results at the local level, further research is recommended to employ bias correction techniques and dynamic downscaling with higher resolution.

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CRedit Author Contribution

L. S. Saadah: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing – original draft preparation, writing – review and editing.

H. Halimurrahman: conceptualization, methodology, validation, supervision, writing – review and editing.

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Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be available on request.

Use of Generative AI and AI-Assisted Technologies

No generative AI or AI-assisted technologies were employed in the preparation of this manuscript.

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